Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The end product (C) in the following sequence of reactions is

$$HC \equiv CH \xrightarrow[20\%\ H_2SO_4]{1\%\ HgSO_4} A \xrightarrow[H_2O]{CH_3MgX} B \xrightarrow{[O]} (C)$$

(A) acetic acid

(B) isopropyl alcohol

(C) acetone

(D) ethanol

Q.2

Predict the product 'B' in the sequence of reactions

$$CH \equiv CH \xrightarrow[\%\ HgSO_4]{30\%\ H_2SO_4} A \xrightarrow{NaOH} B$$

(A) CH_3COONa

(B) CH_3COOH

(C) CH_3CHO

(D) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CHO$

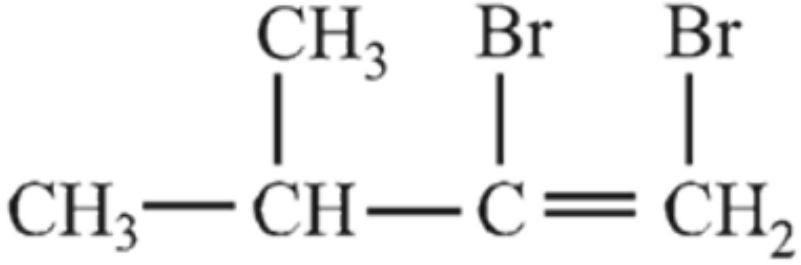
Q.3

$$CH_3-\underset{\underset{CH_3}{|}}{CH}-C\equiv CH \xrightarrow{\text{excess HBr}}$$

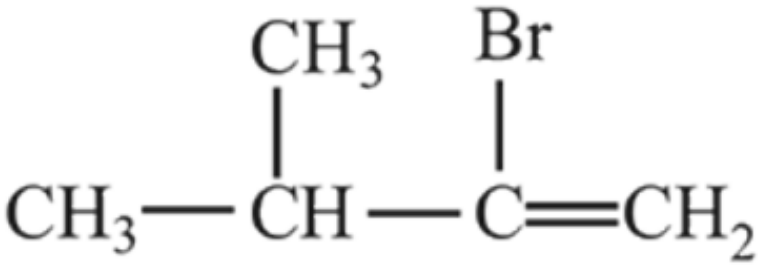
The product of the above reaction is:

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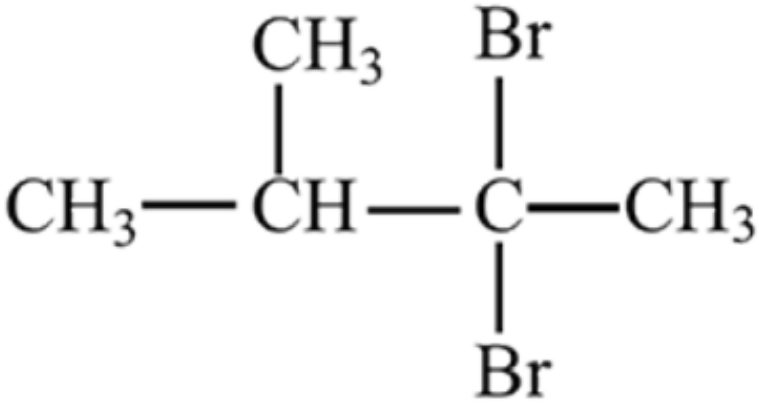
(A)



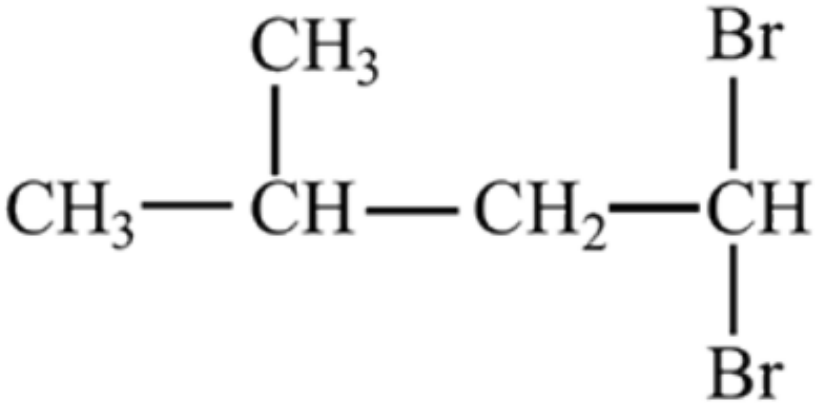
(B)



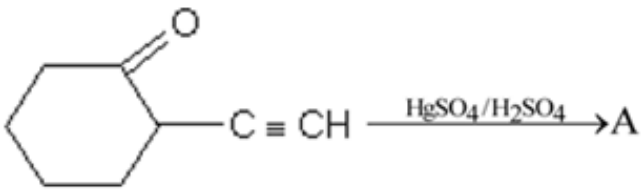
(C)



(D)



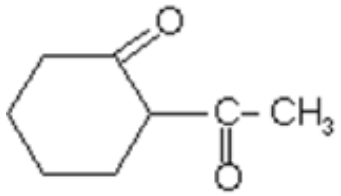
Q.4



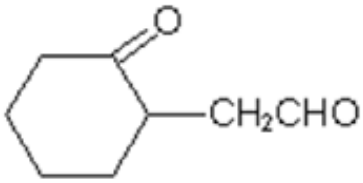
is

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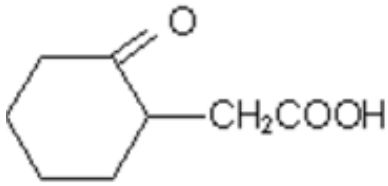
(A)



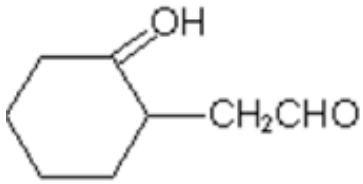
(B)



(C)

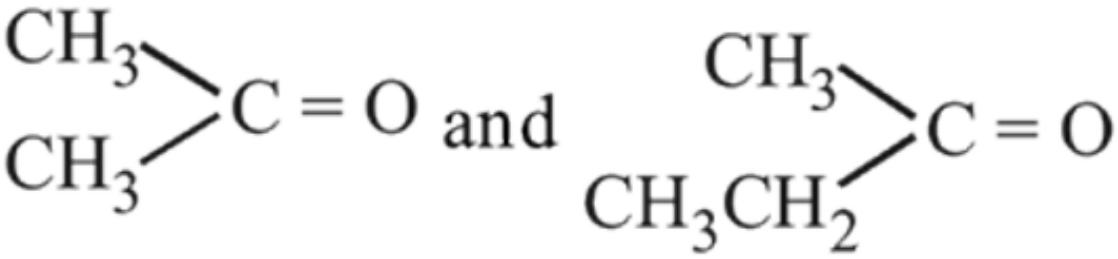


(D)



Q.5

An alkene having molecular formula C_7H_{14} was subjected to ozonolysis in the presence of zinc dust. An equimolar amount of the following two compounds was obtained



The IUPAC name of the alkene is

(A) 3,4 -dimethyl-3-pentene

(B) 3,4 -dimethyl-2-pentene

(C) 2,3-dimethyl-3-pentene

(D) 2,3-dimethyl-2-pentene

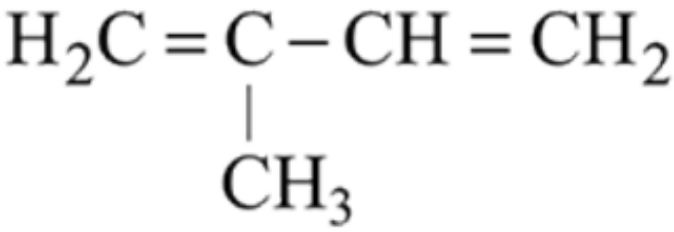
Q.6

Which of the following will yield a mixture of 2-chlorobutene and 3-chlorobutene on treatment with HCl?

(A) $CH_2 = C = CH - CH_3$

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(B)



(C) $CH_2 = CH - CH = CH_2$

(D) $HC \equiv C - CH = CH_2$

Q.7

Which of the following statements is incorrect regarding dehydrohalogenation of alkenes?

(A) During the reaction hydrogen atom is eliminated from the β - carbon atom.

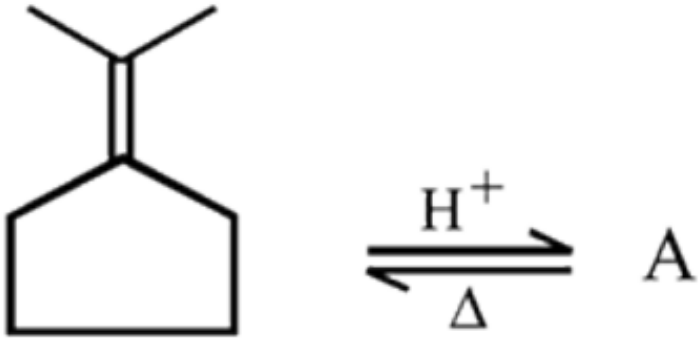
(B) Rate of reaction for same alkyl group; Iodine $>$ Bromine $>$ Chlorine

(C) Rate of reaction; $(CH_3)_3\,C- > (CH_3)_2\,CH- > CH_3CH_2-$

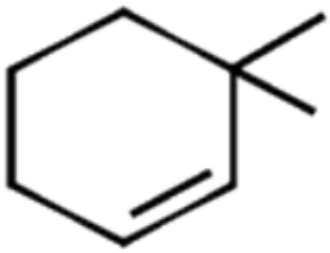
(D) Only nature of halogen atom determine rate of the reaction.

Q.8

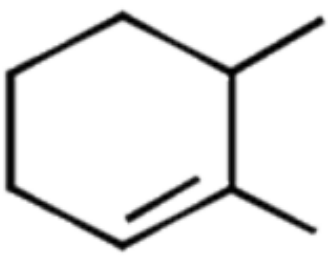
Predict the product (A) of the following reaction



(A)

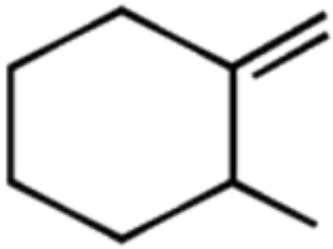


(B)

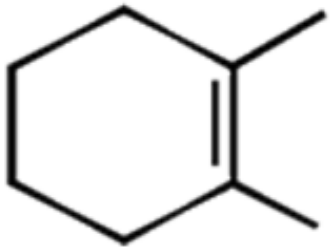


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(C)



(D)



Q.9

In the hydroboration - oxidation reaction of propene with diborane, H_2O_2 and $NaOH$, the organic compound formed is:

(A) CH_3CH_2OH

(B) $CH_3CHOHCH_3$

(C) $CH_3CH_2CH_2OH$

(D) $(CH_3)_3\,COH$

Q.10

Which one of the following compounds would have the highest heat of hydrogenation?

(A) $CH_2 = CH_2$

(B) $CH_3 - CH_2 - CH = CH_2$

(C) $CH_3CH = CHCH_3$

(D) $(CH_3)_2\,C = C\,(CH_3)_2$

Q.11

Which of the following types of reaction occur when a reactant has got a double bond ?
(i) Addition
(ii) Photolysis
(iii) Nucleophilic substitution
(iv) Polymerization

(A) (i) and (iv)

(B) (i) and (ii)

(C) (i), (ii) and (iv)

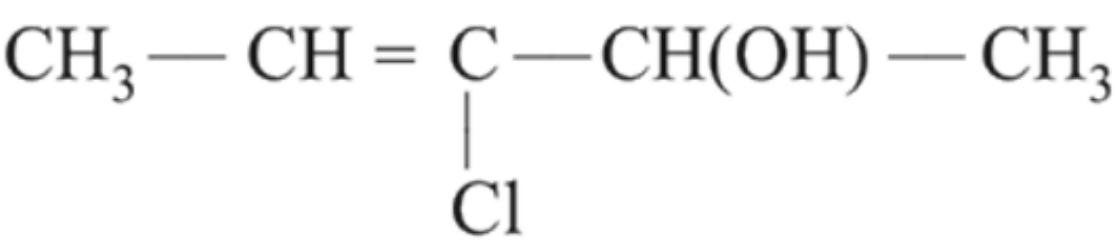
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(D) (i), (ii), (iii) and (iv)

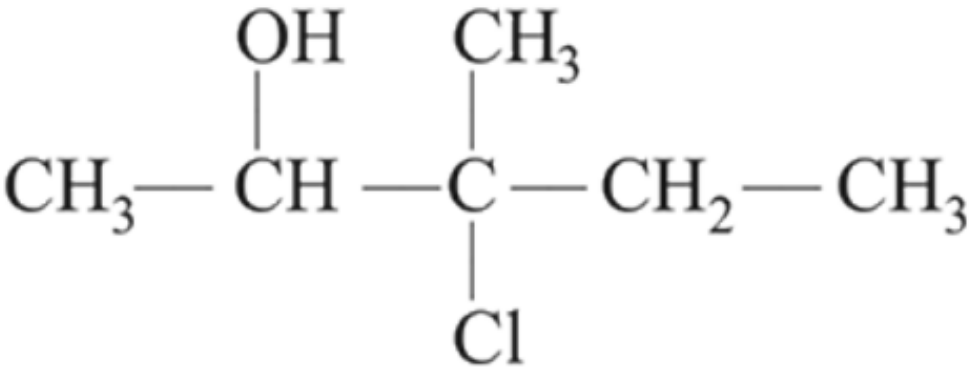
Q.12

$HOCl$ reacts on 3 -methyl-2-pentene, the main product will be :

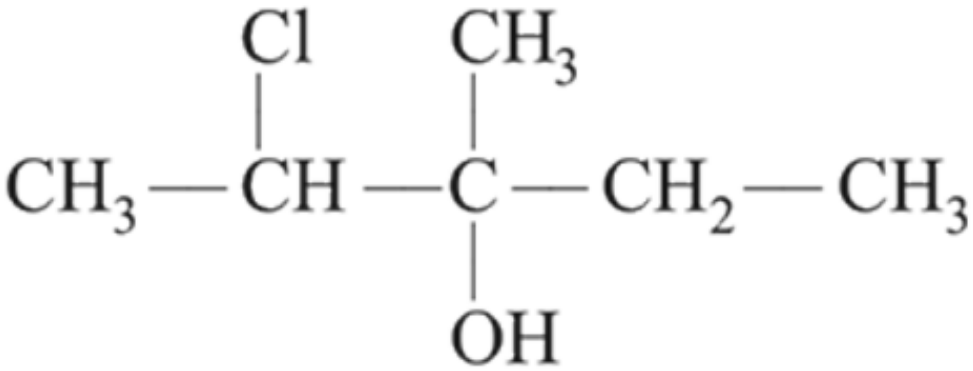
(A)



(B)



(C)



(D) None of these

Q.13

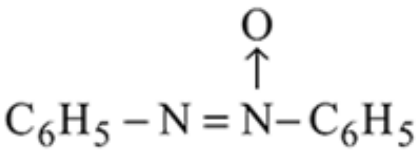
In the acidic reduction of nitrobenzene, which of the following is the intermediate?

(A) $C_6H_5 - N = O$

(B) $C_6H_5NH - NHC_6H_5$

(C) $C_6H_5 - N = N - C_6H_5$

(D)



Q.14

Ethylene can be prepared in good yield by

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(A) $CH_3CH_2\,N^+(CH_3)_3\,I^- \xrightarrow{\text{heat}} CH_2=CH_2 + (CH_3)_3\,N + HI$

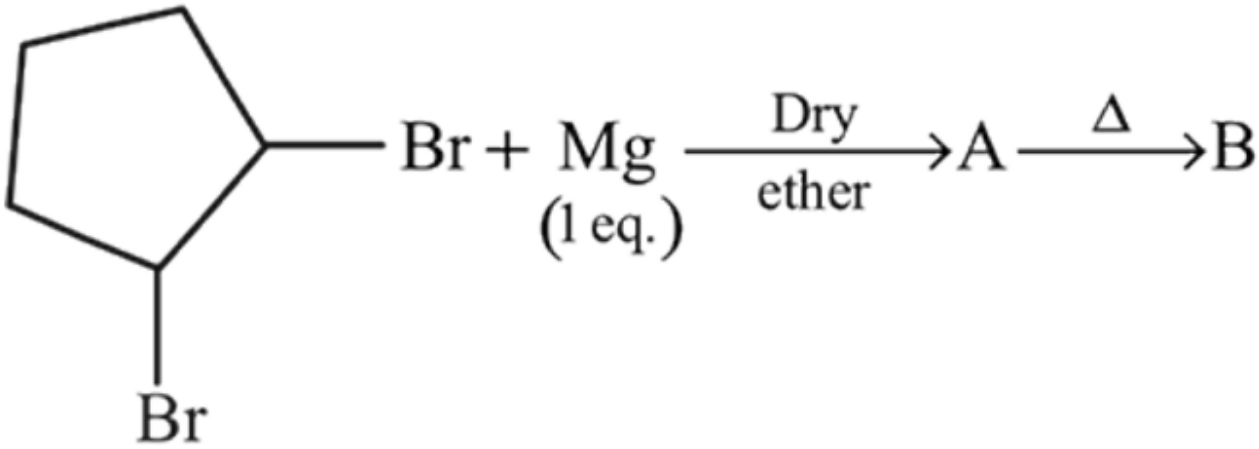
(B) $CH_3CH_2\,N^+(CH_3)_3\,OH^- \xrightarrow{\text{heat}} CH_2=CH_2 + (CH_3)_3\,N + H_2O$

(C) Both (a) and (b)

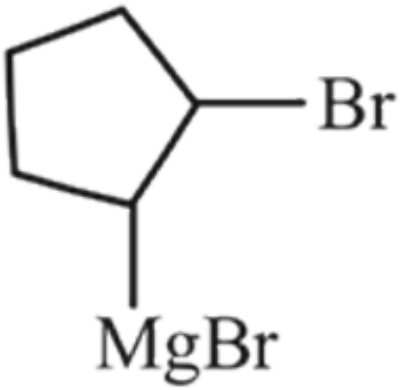
(D) $CH_3CH_2NH_2 \xrightarrow{\text{heat}} CH_2=CH_2 + NH_3$

Q.15

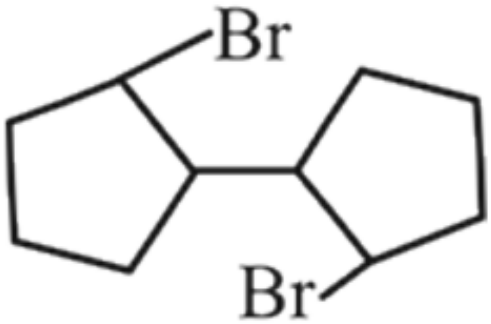
In the following reaction, compound (B) is



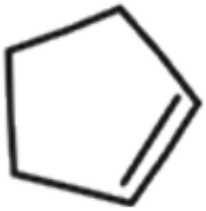
(A)



(B)

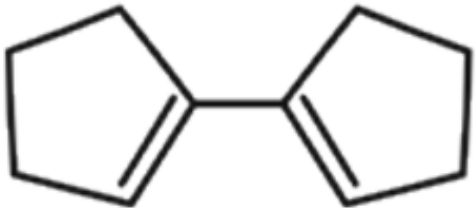


(C)



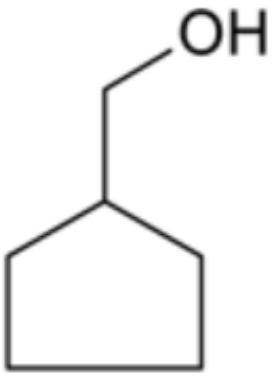
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(D)

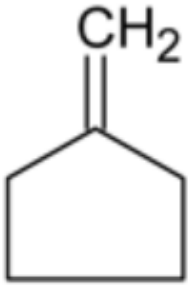


Q.16

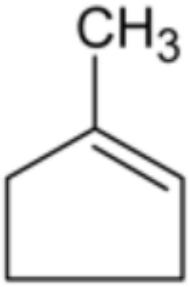
In the dehydration of,



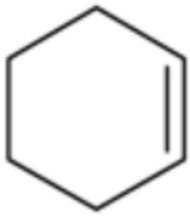
The product formed are



(I)



(II)



(III)

(A) Only I

(B) Only II

(C) Only III

(D) I, Il and II

Q.17

Which of the following wil have least hindered rotation about carbon carbon bond?

(A) Ethane

(B) Ethene

(C) Ethyne

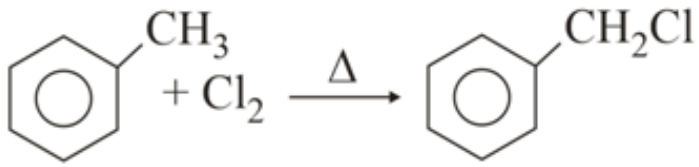
(D) Hexachoroethane

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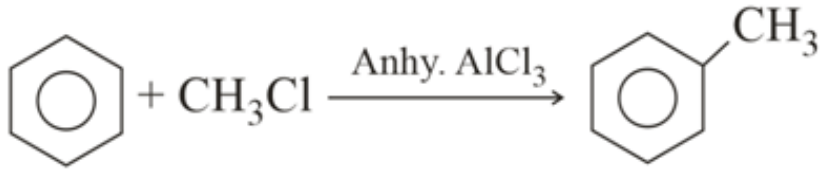
Q.18

Which of the following is a free radical substitution reaction ?

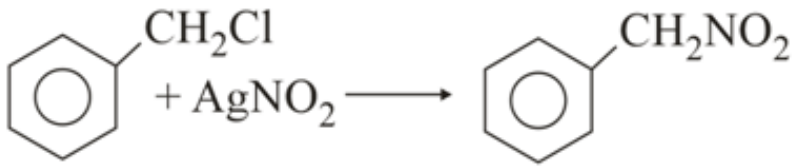
(A)



(B)

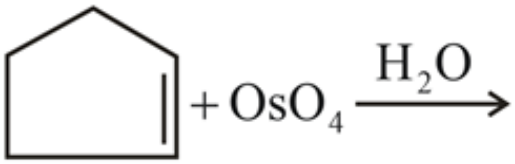


(C)



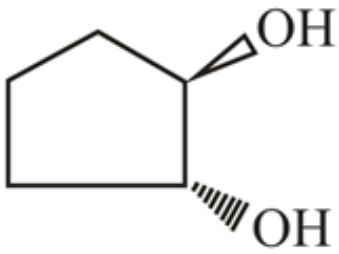
(D) $CH_3 - CHO + HCN \longrightarrow CH_3 - CH(OH)\,CN$

Q.19

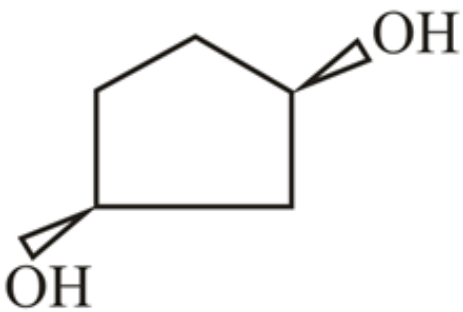


Select major product (s) : -

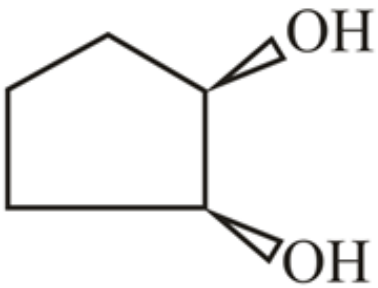
(A)



(B)

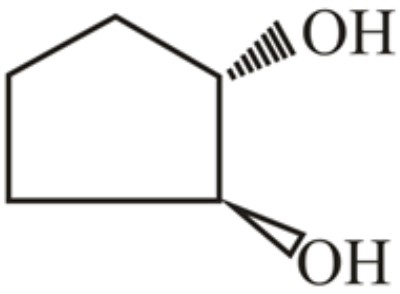


(C)

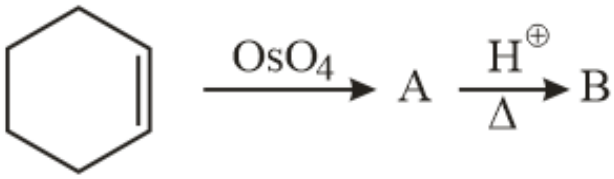


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(D)

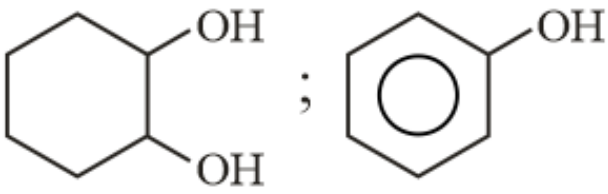


Q.20

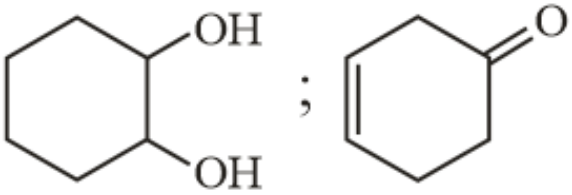


A & B are respectively

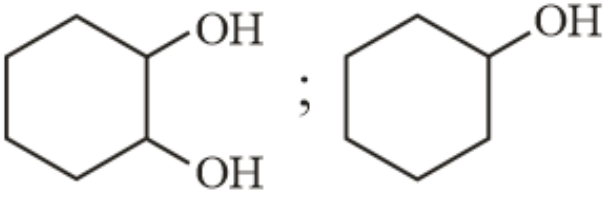
(A)



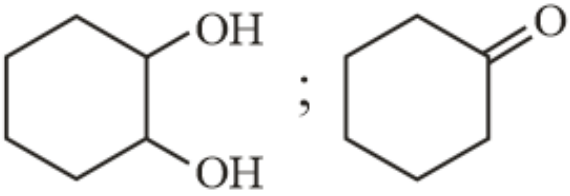
(B)



(C)



(D)

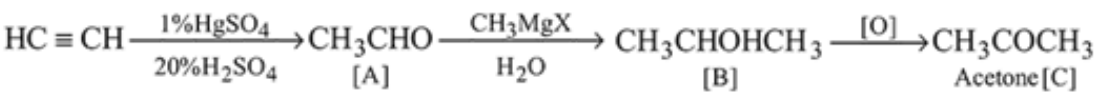


Answers & Solutions

Q.1 Answer:

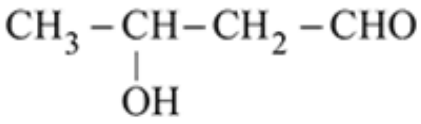
acetone

Solution:

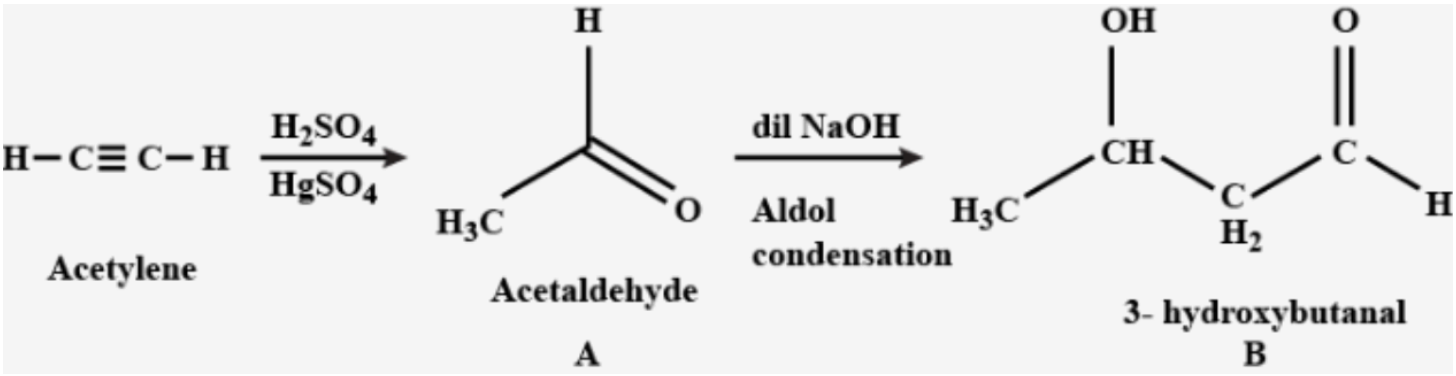


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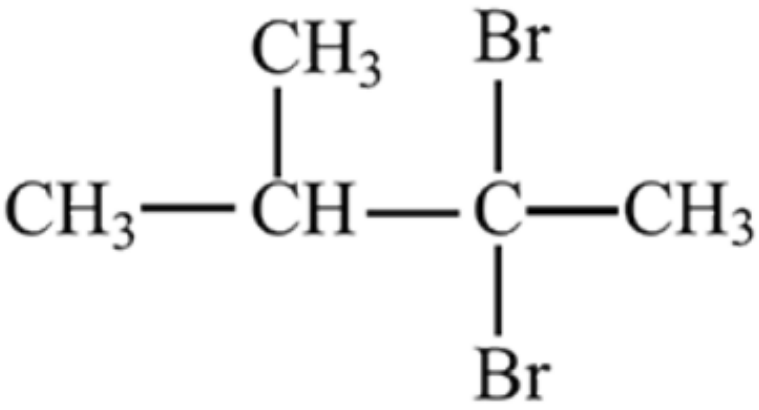
Q.2 Answer:



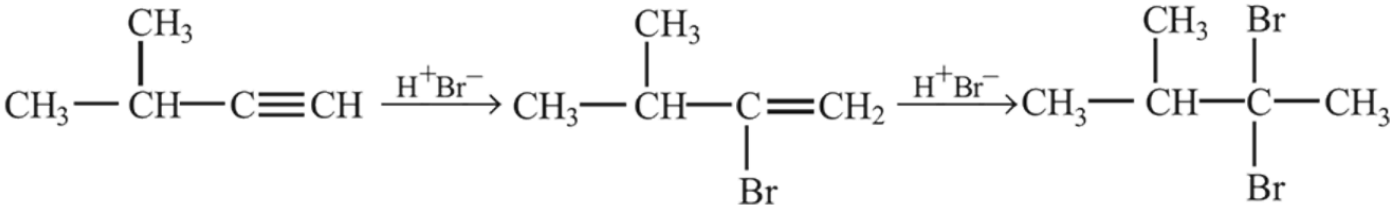
Solution:



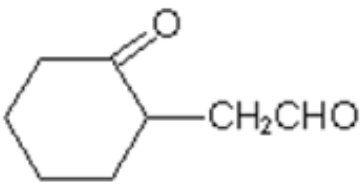
Q.3 Answer:



Solution:

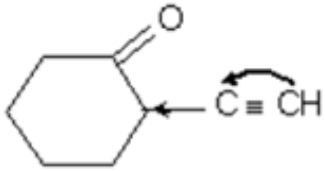


Q.4 Answer:



Solution:

C = O **is (EWG) hence** π **electron transfer is onwards ring.**

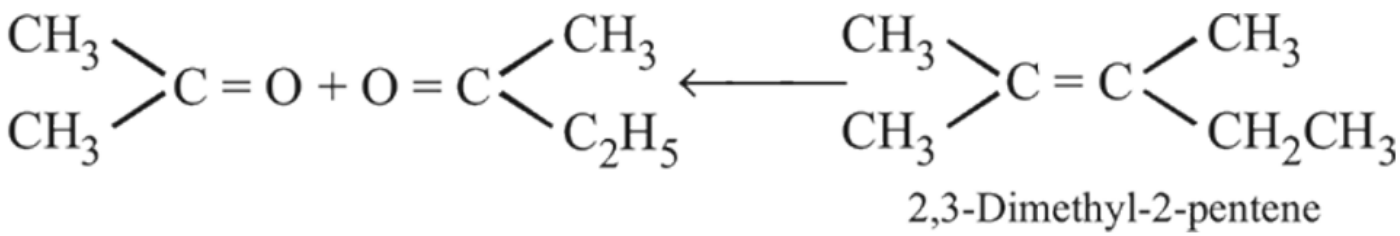


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Q.5 Answer:

2,3-dimethyl-2-pentene

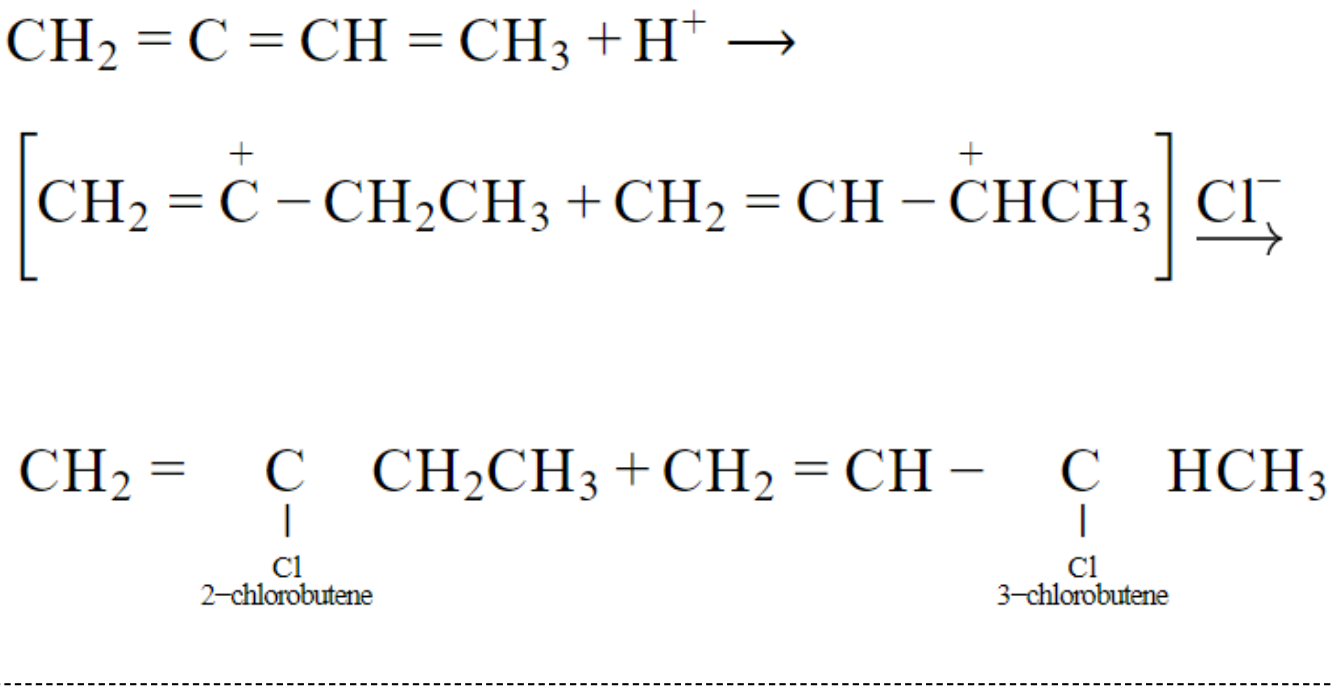
Solution:



Q.6 Answer:

$CH_2 = C = CH - CH_3$

Solution:



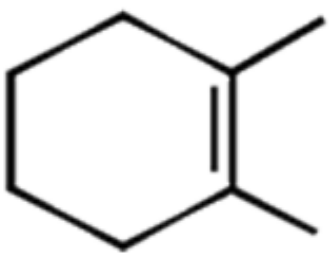
Q.7 Answer:

Only nature of halogen atom determine rate of the reaction.

Solution:

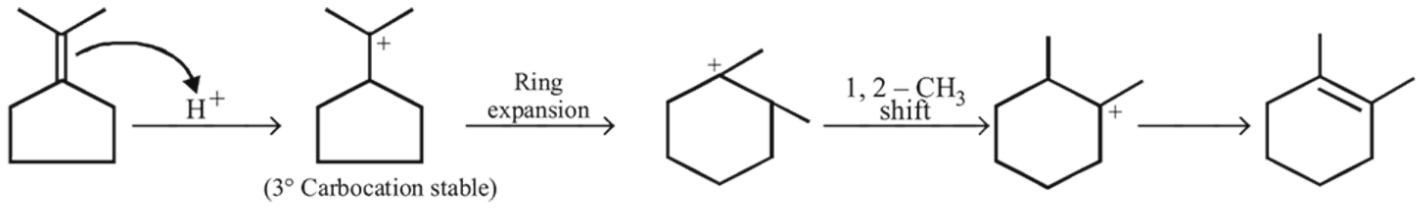
Nature of halogen atom and the alkyl group both determine rate of reaction.

Q.8 Answer:



Solution:

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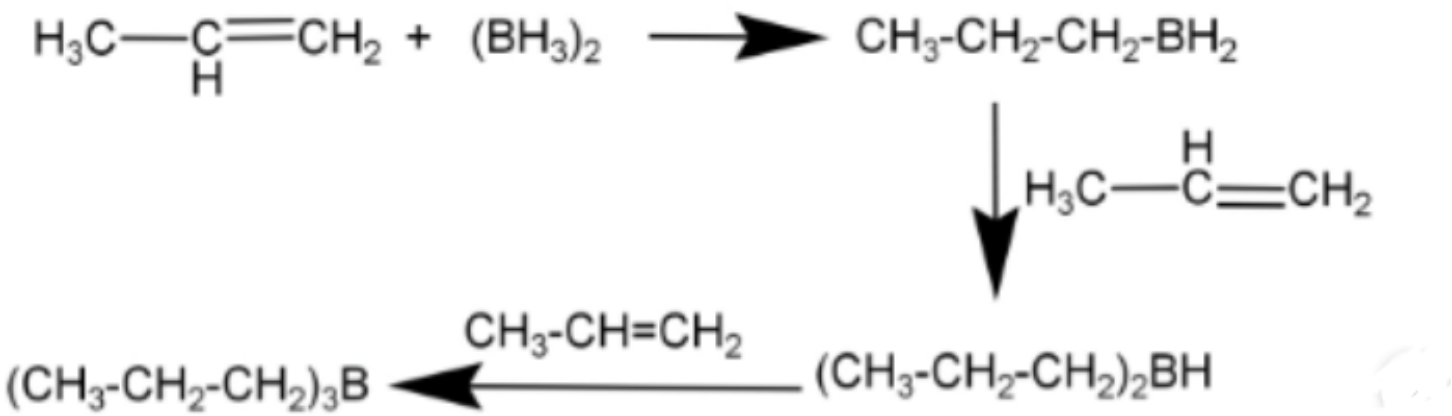


Q.9 Answer:

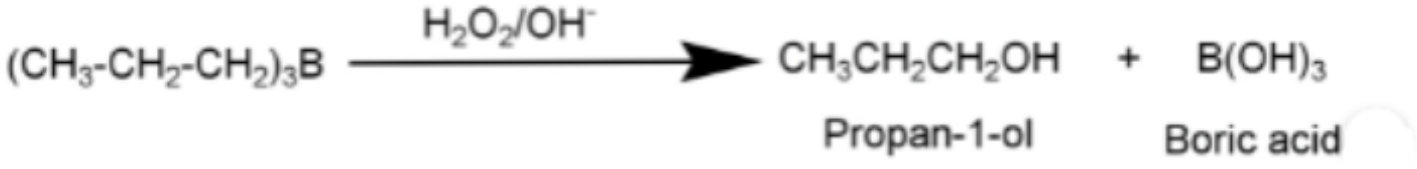
$CH_3CH_2CH_2OH$

Solution:

(1) Hydroboration step: BH_3 reacts with the double bond and transfers one of the hydrogen atoms to the C atom adjacent to the one that bonds with the B atom. This step is repeated two more times until 3 alkyl groups are bonded with the B atom to form R_3B. In case of propene the product formed here is: $(CH_3CH_2CH_2)_3\ \mathrm{B}$



(2) Oxidation step: Now the boron atom is attacked by the hydrogen peroxide ion which acts as a nucleophile and sodium hydroxide (H_2O_2/OH^-) and produces primary alcohol (R-OH). In case of propene the product formed will be $(CH_3CH_2CH_2OH)$.



The final product formed is propan-1-ol $(CH_3CH_2CH_2OH)$.

Q.10 Answer:

$CH_2 = CH_2$

Solution:

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The heat of hydrogenation of an alkene depends upon its stability. Higher the stability, lower the heat of hydrogenation. Since $CH_2 = CH_2$ has no substituent, it is the least stable alkene and hence has the highest heat of hydrogenation

Q.11 Answer:

(i) and (iv)

Solution:

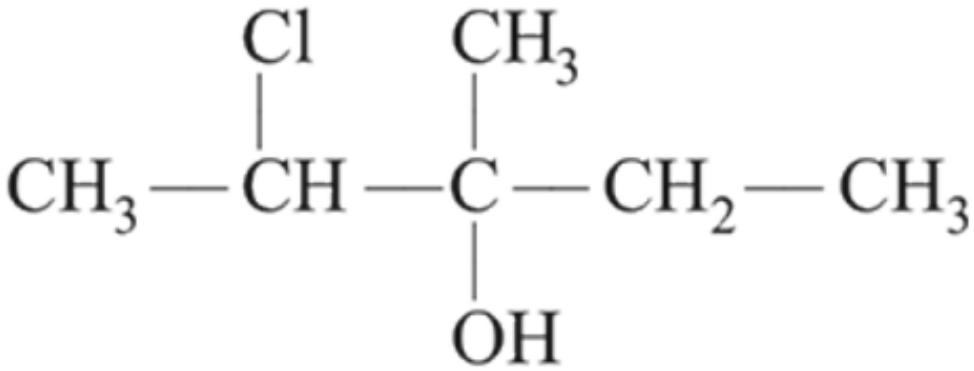
Addition reaction occurs on a double bond. The compound containing double bonds are also undergo polymerisation. So, the correct option are both (i) and (iv) e.g.

$CH_2 = CH_2 + HOCl \longrightarrow CH_2OH - CH_2Cl$

(addition reaction)

$nCH_2 = CH_2 \overset{\text{Polymerisation}}{\longrightarrow} [CH_2 - CH_2]_n$

Q.12 Answer:

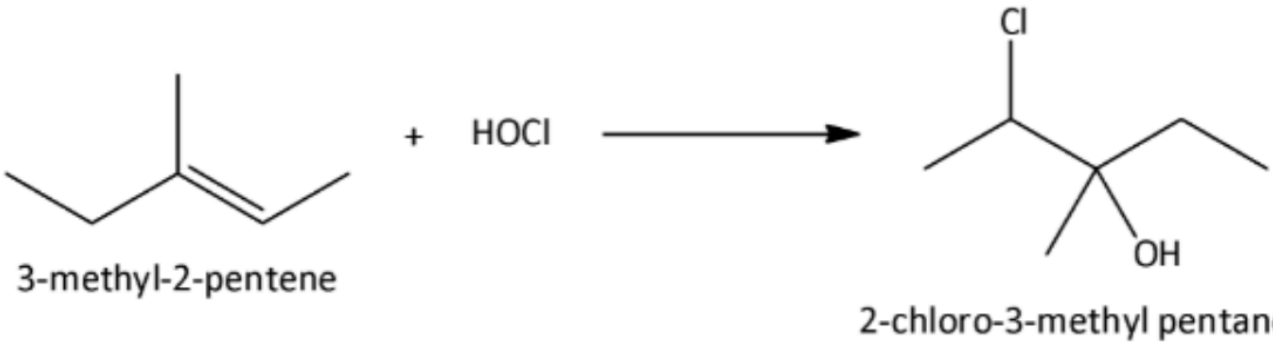


Solution:

When 3-methyl-2-pentene is reacted with hypochlorous acid, the product formed is 2 -chloro-3-methyl pentanol-

3 The reaction obeys Markovnikov's addition and the hydroxide and chloro group is added across the double bond of two carbons.

We can write the reaction of 3 -methyl-2-pentene with $HOCl$ as,



Q.13 Answer:

$C_6H_5 - N = O$

Solution:

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Nitrobenzene is reduced in the following manner through various intermediates to form aniline as the final product.

$$\underset{\text{nitrobenzene}}{C_6H_5NO_2} \xrightarrow{2[H]} \underset{\text{nitrosobenzene}}{C_6H_5NO} \xrightarrow{2[H]} \underset{\text{phenylhydroxylamine}}{C_6H_5NHOH} \xrightarrow{2[H]} \underset{\text{aniline}}{C_6H_5NH_2}$$

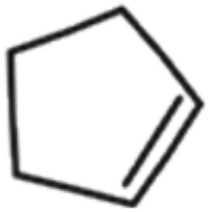
Q.14 Answer:

$$CH_3CH_2\ N^+(CH_3)_3\ OH^- \xrightarrow{\text{heat}} CH_2=CH_2 + (CH_3)_3\ N + H_2O$$

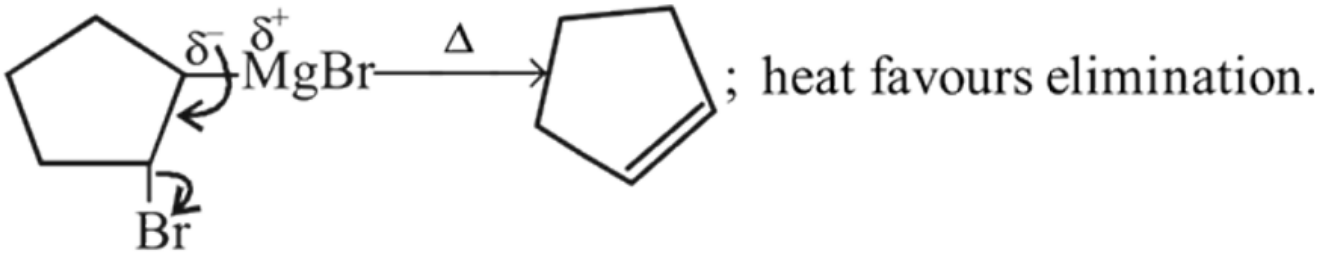
Solution:

This is an example of Hofmann elimination which generally takes place by E2 mechanism and the latter requires a strong base (recall that OH^- is a strong base than I^-). The $NH_2{}^-$, being a strong base can't be eliminiated easily.

Q.15 Answer:



Solution:

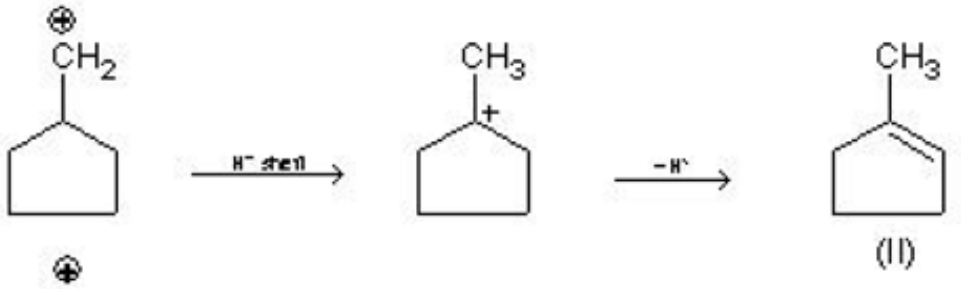


Q.16 Answer:

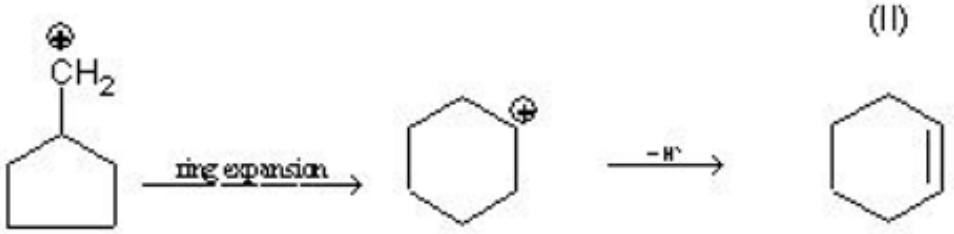
I, Il and II

Solution:

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(II)



(III)

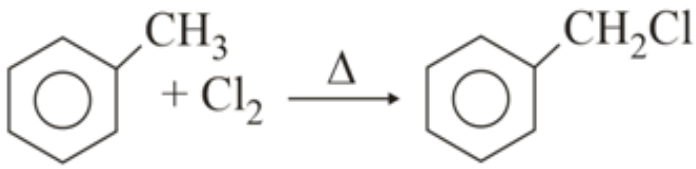
Q.17 Answer:

Ethane

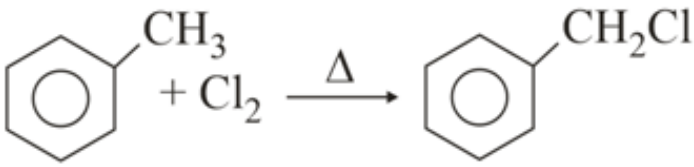
Solution:

Free rotation around carbon-carbon bond takes place easily in alkanes. Now ethane and hexachloroethane both are alkanes, but in hexachloroethane bulky chlorine atom is present while ethane is least hindered.

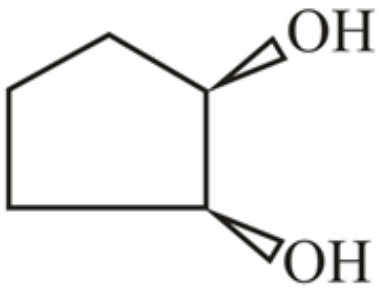
Q.18 Answer:



Solution:



Q.19 Answer:

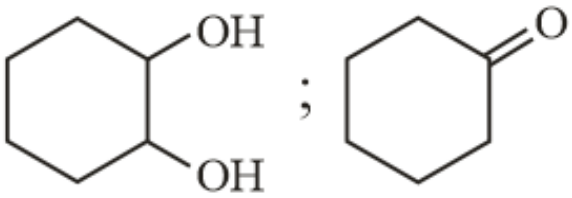


Solution:

OsO_4 **results in syn addition of** 2 "OH" **group.**

Q.20 Answer:

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Solution:

Dihydroxylation reaction of alkene.

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